Robayed Ashraf

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SUMMARY

AI/ML Engineer specialising in production-grade agentic AI, LLM applications, and real-time Computer Vision. Built DocuMind from scratch, an agentic RAG system with hybrid search, async ingestion, multi-user auth, and Contextual Retrieval, achieving 0.984 faithfulness on RAGAS evaluation. Owning everything from pipeline architecture and backend engineering to deployment and CI/CD, and driven by shipping systems that work in the real world.

TECHNICAL SKILLS

AI & Machine Learning: Agentic AI, LLM Applications, RAG, Generative AI, Deep Learning, Computer Vision, NLP, Prompt Engineering, Hybrid Search, Vector Embeddings, Contextual Retrieval, LLM Evaluation (RAGAS)
Frameworks: Python, PyTorch, TensorFlow, Keras, LangChain, LangGraph, FastAPI, OpenCV, Scikit-Learn, HuggingFace, Redis, Celery
MLOps & DevOps: Docker, Docker Compose, GitHub Actions (CI/CD), ClearML, Pytest, REST API Development
Tools & Practices: Git, SQL, PostgreSQL, Agile/Scrum, Jira, Confluence

PROJECTS

DocuMind: Agentic Document Intelligence Feb 26 – Present
Tech: LangGraph, LangChain, Gemini 2.5 Flash, FastAPI, Next.js 16, Postgres, pgvector, Redis, Celery, HuggingFace, Docker, GitHub Actions, RAGAS, Pytest

- An agentic RAG system that lets users chat with PDFs using autonomous query routing, hallucination detection, and self-correcting retry loops powered by LangGraph.
- Hybrid retrieval combining pgvector HNSW dense and PostgreSQL ts_rank sparse search, fused with RRF and cross-encoder reranking, with Contextual Retrieval and HyDE fallback.
- Async Celery ingestion pipeline with five-stage progress tracking, Clerk JWT multi-user auth, Redis semantic cache, and rich PDF parsing with Gemini multimodal figure captioning.
- Achieves 0.984 faithfulness, 0.933 context recall, and 0.882 context precision on a 30-question RAGAS golden dataset; deployed via Docker Compose with GitHub Actions CI/CD.

SignSync: Sign Language Translation for Video Conferencing Feb 25 – Jun 25
Tech: Python, ANN, OpenCV, Gemini API (LLM integration), Computer Vision

- Trained an ANN on 3,200 images across 42 classes for real-time AUSLAN finger spelling classification, achieving 99.79% validation accuracy.
- Integrated Gemini API for LLM-based post-processing, correcting recognition errors, and improving caption readability in real time.
- Designed a 30-frame confidence-based smoothing mechanism to stabilise live predictions in continuous video streams.
- Deployed the full pipeline into a live Jitsi environment, delivering real-time sign-to-text captions as a production-viable accessibility feature.
- Presented at UTS Tech Fest AI Showcase 2025, recognised for practical impact among postgraduate AI projects.

ThirdAxis: Smart Plant Health Monitor Feb 25 – Jun 25
Tech: Python, CNN, ClearML, Streamlit, GitHub Actions, Jira, Confluence

- Managed end-to-end ML project lifecycle using Agile practices with Jira and Confluence, covering data ingestion, preprocessing, training, evaluation, and deployment.
- Designed a modular ClearML pipeline with separate scripts for each stage, enabling reproducible and traceable experimentation.
- Implemented ClearML artifact management for dataset and model versioning, supporting experiment comparison and rollback across training iterations.
- Built a Streamlit inference interface for real-time plant disease classification from leaf images, addressing early disease detection needs in precision agriculture.

Multi-Object Tracking System for Real-Time Video Inference Jan 25 – Jun 25

Tech: PyTorch, OpenCV, YOLOX, MOTR, TrackEval, CUDA, Streamlit

- Compared MOTR (Transformer-based) and BoostTrack++ (tracking-by-detection with YOLOX) on MOT17, with MOTR achieving 98.518% IDF1 and 92.57% HOTA versus 96.104% and 87.093% for BoostTrack++.
- Profiled real-time inference on NVIDIA RTX 4060, recording 1.54 to 3.60 FPS and 255 to 409ms latency to identify deployment trade-offs for live video scenarios.
- Built a modular Streamlit prototype supporting live webcam and uploaded video inference with real-time bounding box overlays, FPS display, and latency tracking.
- Evaluated tracking quality using TrackEval metrics, including MOTA, IDF1, and HOTA, against MOT17 ground truth annotations.

WORK EXPERIENCE

Self-Employed, AI/ML Engineer Feb 26 – Present

- Sole engineer across two major releases, making all architecture decisions from database selection and retrieval design to API structure and deployment strategy.
- Designed and implemented the full backend: FastAPI REST API, async Celery worker queue, Postgres schema with pgvector indexes, Redis caching layer, and Clerk JWT auth integration.
- Built and iterated on the agentic LangGraph pipeline and retrieval stack using RAGAS evaluation metrics to validate every improvement across versions.
- Replaced the Streamlit prototype with a production Next.js 16 frontend featuring SSE streaming chat, an inline PDF citation viewer, and per-user document isolation.

East West University Computer Programming Club, Programming Trainer (Volunteering) Jan 19 – Oct 21

- Mentored 50+ students in algorithms, data structures, and competitive programming.
- Designed and conducted ICPC-style mock contests to build structured problem-solving skills.

EDUCATION

University of Technology Sydney, Master of Artificial Intelligence (Major: Computer Vision) Aug 23 – Jun 25

WAM: 80.94 / 100 (Grade: 6.13 / 7.00)

East West University, Bachelor of Science in Computer Science and Engineering Jan 18 – Oct 21

CGPA: 3.64 / 4.00

- Dean's List recipient and merit-based scholarship awardee for academic excellence.